

Nonlinear Dynamics and Risk Quantification in Nepal Stock Exchange: A Jump-Diffusion COGARCH Approach

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Outline

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Nepal Stock Exchange: Market Profile

About NEPSE

- Established **1993**; Nepal's sole securities exchange.
- **Commercial Banks** sector: $\approx 40\%$ of total market capitalisation — most liquid and economically critical segment [1].
- $T = 2886$ post-break trading days (Oct 2013 – Jun 2026).
- Key features: thin liquidity, retail-dominated, politically sensitive, limited derivatives market.

Why Model This Sector?

- Banking stability is central to Nepal's financial system.
- Existing volatility studies [2] use simple GARCH — insufficient for jump-rich dynamics.

Core Empirical Problem

Standard models assume Gaussian returns and continuous paths. NEPSE banking returns violate both:

- Excess kurtosis = 7.21
- Skewness = 0.90
- 39 jump events over 12 years

Research Question

Do NEPSE banking returns satisfy the near-critical COGARCH conditions for nonlinear dynamics, and what risk estimates follow?

Motivation: Why Do Classical Models Fail?

Facts of Asset Returns [3]

- **Heavy tails:** Gaussian VaR underestimates tail risk.
- **Volatility clustering:** large changes cluster [4].
- **Long memory:** $ACF(|r_t|)$ decays hyperbolically [5].
- **Multifractality:** scaling varies across time scales [6].
- **Jumps:** sudden large moves [7, 8].

Model Evolution

Model	Limitation
Black-Scholes	Constant vol, no jumps
GARCH [9]	Discrete, no jumps
EGARCH [10]	No jumps
Merton jump [7]	Not self-exciting
BN-S SV [11]	No self-excitation
COGARCH [12]	Jump + self-exciting

COGARCH Advantage

Jump-driven **self-exciting volatility feedback**,
direct connection to chaos via branching ratio B .

Research Objectives & Hypotheses

Objectives

- (i) Distributional and chaos diagnostics on NEPSE Commercial Banks daily returns.
- (ii) Estimate COGARCH parameters via the **Method of Simulated Moments** (MSM).
- (iii) Compute B , γ , H , $\Delta\alpha$; compare with theoretical thresholds.
- (iv) Estimate one-year **VaR**, **CVaR**, and **drawdown probabilities** via 10,000-path Monte Carlo.

Falsifiable Hypotheses

- (i) $\gamma \leq 2$: infinite-variance tails
- (ii) $H > 0.5$: long memory in $|r_t|$
- (iii) $\Delta\alpha > 0.1$: multifractal scaling
- (iv) $B \in [0.9, 1)$: near-critical branching

Scope

- **Series**: NEPSE Commercial Banks index
- **Sample**: 1 Oct 2013 - 3 Jun 2026 ($T = 2886$ obs.)
- Break date by Zivot-Andrews test
- Daily frequency; univariate analysis

Literature Review: Volatility & Jump Models

Stochastic volatility & GARCH family

- Engle (1982) ARCH [4]: conditional heteroscedasticity.
- Bollerslev (1986) GARCH(1,1) [9]: persistent clustering.
- Nelson (1991) EGARCH [10]: asymmetric leverage effect.
- Barndorff-Nielsen & Shephard (2001) [11]: non-Gaussian Ornstein-Uhlenbeck stochastic variance.
- Klüppelberg et al. (2004) [12]: continuous-time COGARCH — self-exciting jump volatility.

Jump models & stylised facts

- Merton (1976) [7]: first jump-diffusion option pricing model.
- Cont (2001) [3]: definitive survey of asset return stylised facts — heavy tails, clustering, long memory.
- Duong & Swanson (2010) [8]: empirical evidence of jump persistence.

Gap in NEPSE Literature

Existing work [2] stops at plain GARCH — no jump components, no chaos diagnostics. **This study bridges that gap.**

Fractal geometry & long memory

- Mandelbrot (2013) [13]: fractal scaling and discontinuous price trajectories.
- Peters (1994) [5]: Fractal Market Hypothesis — $H > 0.5$ implies persistent long memory.
- Calvet & Fisher (2002) [6]: multifractal model (MMAR); spectrum width $\Delta\alpha$ measures volatility complexity.

Criticality in financial markets

- Hardiman et al. (2013) [14]: US/UK markets operate near $B \approx 1$ — endogenous self-excitation dominates. Emerging markets: untested.
- World Bank (2020) [1]: Nepal financial sector stability and structural shocks.

Theoretical Bridge

Near-critical COGARCH ($B \rightarrow 1^-$) produces the same signatures as deterministic chaos: $\gamma \leq 2$, $H > 0.5$, $\Delta\alpha > 0.1$ [12, 14].

Data Description & Distributional Diagnostics

Descriptive Statistics

Post-break: Oct 2013 - Jun 2026, $T = 2886$ obs.

Statistic	Value
Mean (daily)	0.0424%
Std. dev. (daily)	1.5321%
Skewness	0.9029
Excess kurtosis	7.2073
Jarque-Bera stat	7,981.75
JB p -value	< 0.0001
Shapiro-Wilk stat	0.8919
SW p -value	< 0.0001

All normality tests **overwhelmingly rejected**
($p < 0.01\%$) — jumps are structural, not outliers [3].

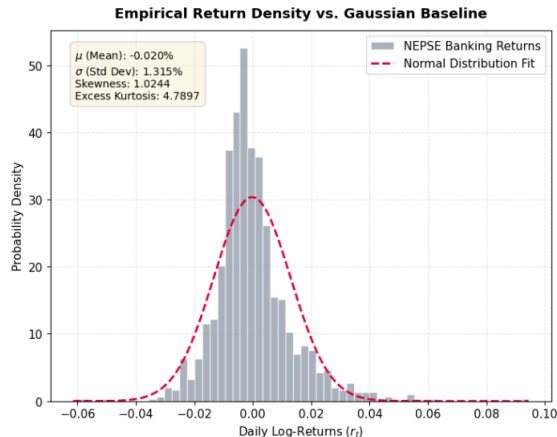
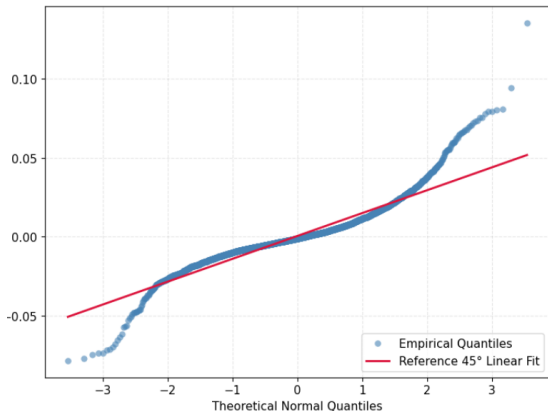


Figure: Empirical density vs. Gaussian: leptokurtic peak and fat tails [3] (Fig. 2.1).

Normal Q-Q Plot & Volatility Clustering

Normal Q-Q Plot (Structural Tail Deviations)



Volatility Clustering Visual Inspection (NEPSE Banking)

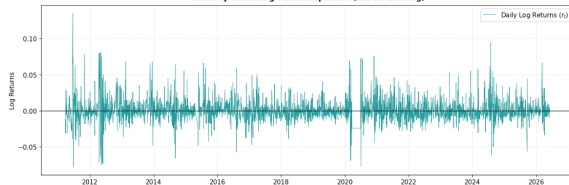


Figure: Log-returns with 21-day rolling volatility corridor: alternating calm and turbulent episodes (Fig. 2.4).

Figure: Normal Q-Q plot: extreme quantiles lie far outside Gaussian confidence bands (Fig. 2.2).

Serial Dependence & ARCH Effects

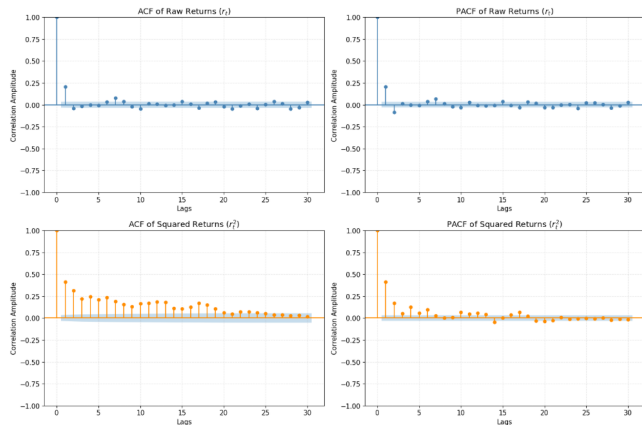


Figure: ACF/PACF: raw (top) and squared returns (bottom) — hyperbolic decay in squared returns signals long-memory clustering [4] (Fig. 2.3).

Test	p -value
LB(10) raw	$< .001$
LB(20) raw	$< .001$
LB(10) sq.	$< .001$
LB(20) sq.	$< .001$
ARCH LM(10)	$< .001$
ADF	$< 10^{-20}$
KPSS	> 0.1

- Strong ARCH effects confirmed.
- **No leverage:** symmetric GARCH admissible.
- **Stationary** series (ADF rejects unit root; KPSS does not reject stationarity).

Jump Detection & Structural Break

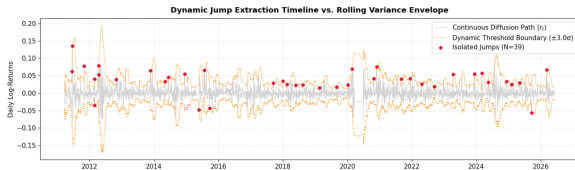


Figure: Dynamic 3σ filter (21-day rolling window): 39 detected jumps, 1.12% of trading days (Fig. 2.5).



Figure: Zivot–Andrews test: endogenous break at **30 Sep 2013** defines homogeneous post-break sample (Fig. 2.6).

Jump Statistics

$\lambda_{\text{annual}} = 2.83$ jumps/year; mean inter-arrival = 88.4 days (median 75.5 days).

Structural Break

ZA stat = -3.17 ($p = 0.852$): no permanent break. Break date used to define a homogeneous estimation window.

Empirical Chaos Diagnostics: Tail Index & Long Memory

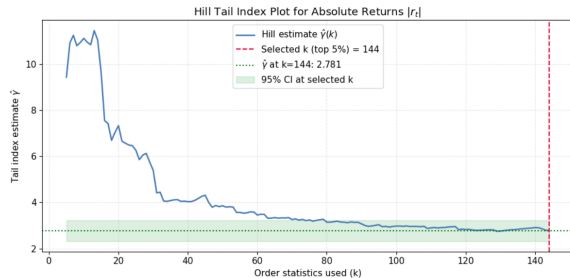


Figure: Hill tail-index $\hat{\gamma}$ vs. order statistics k ; stabilises near $k \approx 140$ (5% of sample) (Fig. 2.7).

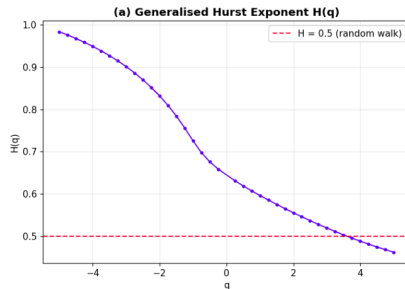


Figure: Generalised Hurst exponent $H(q)$; decreasing profile confirms multifractality [5] (Fig. 2.8).

Tail Index

$\hat{\gamma} = 2.7805 \pm 0.4542$ — finite variance ($\gamma > 2$), power-law tails.

Long Memory

$H = 0.7751 > 0.5$ — persistent dependence in $|r_t|$ [5]. Hyperbolic ACF decay.

Empirical Chaos Diagnostics: Multifractality

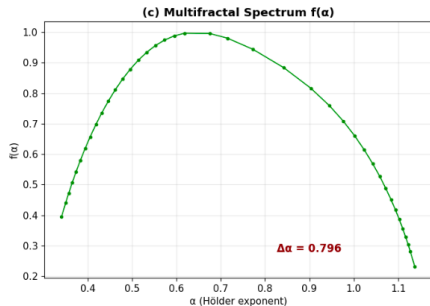


Figure: Multifractal spectrum $f(\alpha)$ vs. Hölder exponent α : broad parabolic shape (Fig. 2.9) [6].

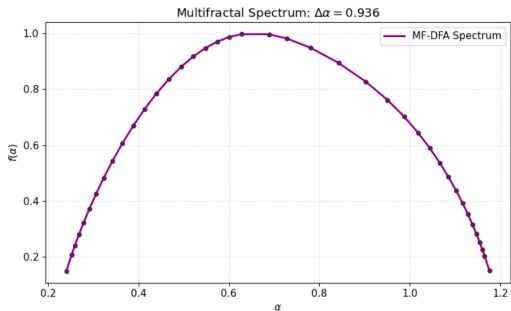


Figure: MF-DFA scaling: non-linear $H(q)$ across moment orders $q \in [-5, 5]$ [6].

Multifractal Width

$\Delta\alpha = 0.7959$ — rich cascade structure consistent with [13, 6].

Implication

Multiple scaling regimes confirm nonlinear, non-normal dynamics consistent with the COGARCH framework.

Model Specification (Klüppelberg, Lindner & Maller, 2004)

Jump-Diffusion Price & Variance SDEs [12]:

$$\frac{dS_t}{S_{t-}} = \alpha dt + \sqrt{V_{t-}} dW_t + (e^{\sqrt{V_{t-}}Z} - 1) dN_t,$$
$$dV_t = \kappa(\theta - V_{t-}) dt + \phi V_{t-} Z^2 dN_t,$$

with W_t Brownian motion, $N_t \sim \text{Poisson}(\lambda)$,
 $Z \sim \mathcal{N}(0, 1)$ i.i.d.

Branching ratio:

$$B = \frac{\lambda \phi \mathbb{E}[Z^2]}{\kappa}$$

$B \rightarrow 1^-$: near-critical self-excitation. $B < 1$: stable mean-reverting volatility.

GARCH(1,1) Discrete Approximation

$$\omega = \kappa\theta,$$

$$\alpha_1 = \phi\lambda,$$

$$\beta_1 = 1 - \kappa.$$

Persistence: $\alpha_1 + \beta_1 = 1 - \kappa + \phi\lambda$.

Key Symbols

κ : mean reversion; θ : long-run variance; ϕ : jump feedback; λ : jump intensity. See List of Symbols (thesis front matter).

Method of Simulated Moments (MSM) & Estimation Results

MSM procedure [12]: minimise weighted squared distance between empirical and simulated moments:

- Mean, std. dev., skewness, excess kurtosis
- ACF of $|r_t|$ at lags 1 and 5
- Hill tail index (top 5% of $|r_t|$)

Optimiser: **differential evolution** (10 paths per evaluation).

Parameter	Estimate
κ (annual)	0.0729
θ (annual var.)	0.000213
ϕ	0.7100
λ (daily)	0.0767
λ (annual)	19.335
Branching ratio B	0.7468

Branching Ratio

$$B = \frac{0.0767 \times 0.7100}{0.0729} = 0.7468$$

Below the near-critical window $[0.9, 1)$ documented in developed markets [14].

GARCH(1,1) Persistence

$$\alpha_1 = 0.0545, \quad \beta_1 = 0.9271$$

$$\alpha_1 + \beta_1 = \mathbf{0.9815}$$

Very high persistence confirms strong volatility clustering.

Model-Implied vs. Empirical Diagnostics

Table: Empirical vs. model-implied chaos diagnostics — synthetic path of 50,000 observations (Table 4.1 in thesis)

Diagnostic	Empirical	Model-implied	Match?
Tail index γ (Hill, $ r_t $)	2.7805	2.8928	✓
Hurst exponent H (DFA, $ r_t $)	0.7751	0.8260	✓
Multifractal width $\Delta\alpha$	0.7959	0.8096	✓

The slight overestimation of H and $\Delta\alpha$ is within sampling variability. The COGARCH captures tail heaviness, long memory, and multifractal structure remarkably well.

Chaotic Conditions Check & Filtered Volatility

Theoretical Conditions [12]

- (i) Near-critical: $B \in [0.9, 1)$? **No** — $B = 0.7468$ ×
- (ii) Heavy tails: $\gamma \leq 2$? **No** — $\gamma = 2.78$ ×
- (iii) Long memory: $H > 0.5$? **Yes** — $H = 0.775$ ✓
- (iv) Multifractality: $\Delta\alpha > 0.1$? **Yes** — $\Delta\alpha = 0.796$ ✓

Two of four conditions met. NEPSE does *not* exhibit the **full** chaotic regime, but long memory and multifractality **alone** justify the COGARCH for risk management.

Why $B < 0.9$? [14]

- Lower liquidity in NEPSE than in developed markets.
- Fewer algorithmic / HFT participants.
- Smaller derivatives market — weaker self-exciting feedback loop.

GARCH Filtered Volatility

Persistence $\alpha_1 + \beta_1 = 0.9815$ confirms strong clustering. Time-varying conditional variance detects more jump days than the simple rolling filter.

Monte Carlo Simulation: Price Paths & Return Distribution

(a) Simulated Price Paths (100 of 10,000)

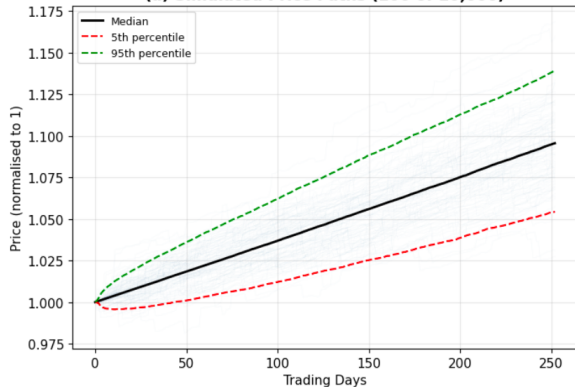


Figure: 100 of 10,000 COGARCH paths (252 days). Black: median; red: 5th pctile; green: 95th pctile. Fan shape reflects volatility persistence and jump risk (Fig. 5.1).

(b) 1-Year Return Distribution

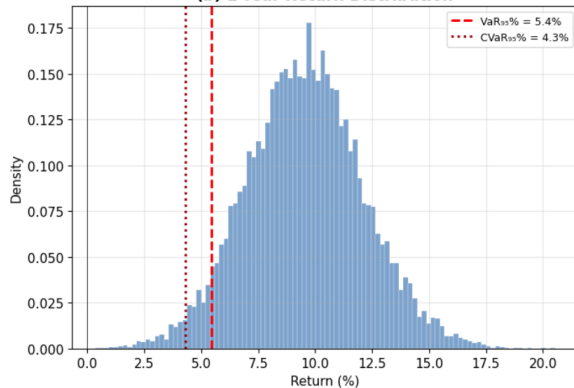


Figure: Terminal 1-year simple return distribution: right-skewed, mode near +10%, small heavy left tail (Fig. 5.2).

Risk Measures & Gaussian Comparison

One-year Risk Measures (10,000 COGARCH paths, Table 5.1 in thesis)

Measure	Value
$\text{VaR}_{95\%}$	-1.7%
$\text{VaR}_{99\%}$	-6.2%
$\text{CVaR}_{95\%}$ (Exp. Shortfall)	-4.4%
$\text{CVaR}_{99\%}$	-8.3%
P(negative annual return)	8.4%
P(drawdown > 30%)	0.0%

Gaussian Benchmark (Ch. 5, Sec. 2)

Annualised vol $\approx 24.3\%$.

Gaussian $\text{VaR}_{95\%} \approx -40\%$ — **grossly over-conservative**.

Gaussian $\text{VaR}_{99\%} \approx -56.5\%$.

COGARCH Advantage

- Mean reversion + jump dynamics $\Rightarrow$ **realistic, not inflated** tail estimates.
- Daily $\text{VaR}_{95\%} \approx -0.11\%$: operationally usable.
- $\text{CVaR} \gg \text{VaR}$: tail losses are severe when they occur — critical for capital adequacy rules.

Discussion: Interpretation of Results

- All EDA evidence (density, Q-Q, ACF, multifractal spectrum) points to **nonlinear, non-equilibrium dynamics** in the NEPSE banking sector.
- Estimated COGARCH parameters reproduce empirical γ , H , $\Delta\alpha$ within sampling variability.
- $B = 0.7468$ — self-exciting feedback **weaker** than in US/UK markets [14]: low liquidity, few HFT participants, small derivatives market.
- $\gamma \approx 2.78$ implies **finite variance** ($\gamma > 2$) — less extreme than crisis-era developed markets, yet close to the critical value $\gamma = 2$.

For Investors & Risk Managers

- COGARCH VaR/CVaR far more accurate than Gaussian models.
- Annual jump intensity = 19.3/year: frequent volatility spikes relevant for position sizing and stop-loss rules.
- 8.4% probability of negative annual return: useful benchmark for strategic asset allocation.

For Regulators

- Use COGARCH to stress-test under simulated jump scenarios.
- Branching ratio B as an early-warning metric for systemic criticality in the banking sector.

Discussion: Limitations & Future Work (Ch. 6, Sec. 3)

- ① **Data frequency:** Daily data misses intraday jumps; 5-min or 1-min data would sharpen λ and ϕ estimates.
- ② **Constant jump intensity:** Poisson λ is restrictive; a Hawkes self-exciting process [14] would capture jump clustering.
- ③ **Parameter stability:** No permanent break found (ZA test), but a regime-switching COGARCH via rolling-window MSM could reveal time-varying dynamics.
- ④ **Model comparison:** Future work should compare COGARCH with GARCH-jump and EGARCH using AIC/BIC and out-of-sample forecasting.
- ⑤ **Multivariate extension:** Banking sector index is an aggregate; systemic risk across individual banks requires a multivariate COGARCH.

- **EDA** (Ch. 2): NEPSE returns exhibit non-normality, volatility clustering [9], long memory ($H = 0.775$) [5], and multifractality ($\Delta\alpha = 0.796$) [6].
- **COGARCH estimation** (Ch. 3): MSM yields $B = 0.7468$, persistence = 0.9815, $\lambda_{\text{annual}} = 19.3$ [12].
- **Chaos validation** (Ch. 4): model reproduces empirical γ , H , $\Delta\alpha$; 2/4 criticality conditions met.
- **Risk quantification** (Ch. 5): $\text{VaR}_{95\%} = -1.7\%$, $\text{CVaR}_{95\%} = -4.4\%$, $\text{VaR}_{99\%} = -6.2\%$ (one-year) vs. Gaussian over-estimate of $\approx -40\%$.
- Provides a **replicable blueprint** for testing nonlinear criticality in any emerging equity market [1, 2].



Thank You for Your Attention!

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Questions?

Please feel free to ask.

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